Edited by Mohsen Ziai, M.D.*

A Potassium Problem

Contributed by Robert E. Cooke, M.D.**

AN-18-MONTH OLD girl was admitted because of diarrhea. She was a well-nourished child with mild dehydration. Serum electrolyte concentrations indicated a slight degree of acidosis with no disturbance in sodium concentration, indicating isotonic dehydration.

The resident staff caring for the child felt that she was so minimally dehydrated that she did not need intravenous fluids. Instead, subcutaneous Darrow's K-lactate and glucose were given. Approximately two hours later, the senior resident was called by the interns and junior residents because the child was obviously deteriorating. Circulation seemed reduced. The cardiac rate was 100, but sounds seemed to be of poor quality. The question was raised whether or not the patient was suffering from potassium deficiency or excess.

An electrocardiogram was taken and immediately indicated the high-peaked T waves of potassium intoxication. The administration of glucose and extracellular electrolytes—sodium chloride and bicarbonate—rapidly corrected the cardiac abnormality with striking clinical improvement.

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Comment

This case illustrates that electrolytes of the intracellular type should not be given before plasma volume is expanded, and that it is most important in the therapy of diarrhea to restore extracellular volume and thereby improve the circulation in critical organs including the kidney.

The differentiation between potassium excess or potassium deficiency is not easy. Bradycardia or tachycardia may occur in both, even though, more typically, bradycardia is a result of potassium intoxication. Serum electrolyte concentrations may be somewhat deceptive as regards potassium deficiency, since, in the presence of acidosis and some reduction in perfusion of the body as in dehydration, serum potassium may seem elevated but may fall rapidly on rehydration, particularly when the acidosis is corrected rapidly.

In general, cardiac function, smooth muscle function, and skeletal muscle function are related to the rate of percentage change of concentration rather than any absolute level. Small changes in extracellular concentration may produce very large changes percentage-wise, as contrasted with fairly large changes in intracellular concentration which may produce relatively small changes in the gradient across the membranes.

Edited by Mohsen Ziai, M.D.*

Adrenal Insufficiency

Contributed by Robert E. Cooke, M.D.**

A 40-YEAR-OLD Addisonian was admitted to the medical service with dehydration. He was given rather large amounts of 5 per cent glucose and water subcutaneously. This led to rapid deterioration in his clinical condition and the development of shock and anuria.

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Comment

It is worth remembering that 5 per cent glucose in water without electrolyte administered anywhere except in the vascular compartment can lead to significant shifts in the extracellular volume. The migration of sodium ions into the new pool of fluid is much more rapid than the migration of glucose particles out, because of the larger size of the glucose particles. Water moves along with the sodium, thus further depleting the vascular compartment temporarily. Hypovolemia is pro-

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duced and its consequence is shock and anuria. The situation was even more acute in this particular case because of the presence of adrenal cortical insufficiency with marginal volume of extracellular fluid and marginal sodium concentration.

One of the symptoms of acute reduction in sodium concentration is apnea. Although this is more often seen in infants, I have observed it in a 30-year-old woman who had extensive pelvic surgery and who developed acute postoperative water intoxication from excessive amounts of fluid given and an inappropriate ADH response. In general, the degree of surgical intervention and trauma to tissue determines the extent of the sequestration and the degree of ADH response but it is difficult to quantitate this particular relationship.

Edited by Mohsen Ziai, M.D.*

Vasculitis with Renal Insufficiency

Contributed by Robert E. Cooke, M.D.**

THIS FOUR-YEAR-OLD girl had a urinary tract infection which was treated with sulfamethoxazole (Gantanol) several months ago. When the infection recurred, she was again treated, a few weeks before admission, with the same drug in the usual doses. Approximately one week after the resumption of this therapy, she developed a rash which was thought possibly to be scarlatinal in origin with desquamation. This rash was accompanied by spiking fever, definite muscle wasting, and a rather striking thickening of the skin of the hands and feet. Prednisone treatment, which was given for several days, was discontinued prior to her transfer to the hospital here. Following the discontinuance of steroids, she became somewhat worse with the development of anuria and a rising urea nitrogen level up to 120 mg/100 ml.

She had no obvious signs of dehydration. Blood pressure was not elevated. The therapeutic problem was not solely one of fluid therapy, although the decision had to be made as to whether or not: 1) she had a prerenal deviation of fluid with oliguria and anuria as a result; 2) she had had some intoxication producing tubular necrosis; and 3) she had some extreme vascular disease as part of the hypersensitivity reaction. She had not had the sulfonamide drug for approximately three weeks by the time she was admitted to the hospital.

No evidence of prerenal deviation could be found. Blood pressure was well sustained. Peripheral circulation seemed good and she had no signs of dehydration.

She was treated as if she had an acute glomerulonephritis, with a high sugar intake and a water intake limited to extrarenal losses. She was given steroids because it was felt that this was a hypersensitivity reaction.

With the administration of steroids in large doses, the urine output progressively increased. The blood urea level did not fall rapidly even though her creatinine levels did. This dissociation between blood urea and creatinine is common with steroid therapy. A striking mobilization of nitrogen from glyconeogenesis and increased protein catabolism occurs, and this leads to an increased liberation of urea at a time when the creatinine levels may be falling as renal function improves.

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